ASSESSMENT REMIT

Name			
ID#			
Semester	Fall 2026	**Section**	
School	School of Engineering, Architecture and Interior Design (EAID)		
Program	Bachelor of Science In Computer Science/ Bachelor of Science in Information Technology		
Major	Computer Science/Information Technology		
Course Code	**CSCI 300**		
Course Title	Operating Systems		
Level	☐ Remedial ☐ Bridging ☑ Undergraduate ☐ Graduate		
Assessment Title	AI-Assisted Operating Systems Application Development and Case Presentation		
Weighting	15 Marks		
Instructor	Dr. Maher Alrahhal		
Hand Out Date	27/10/2026		
Due Date & Time	Date: 15/12/2026	Time: 07:00 – 09:00 PM	
Feedback Post Date	22/12/2026		
Assessment Mode	☐ Individual ☑ Group ☐ Closed Book ☑ Open Book		
Assignment Length *(for assignment-based assessments)*	Exactly 5 pages of documentation, excluding presentation slides and code		
Submission Mode	☐ Online by Email ☑ Online through LMS | TurnItIn ☐ Hard Copy		
Document Type	☐ Words ☑ PDF ☐ Artwork ☑ Software ☑ Presentation		

GENERAL REQUIREMENTS

- The assessment must be completed in groups of four students.
- Each group must develop a functional, interactive Operating Systems application using VS Code with Codex, Claude, or Google Antigravity.
- The application must demonstrate resource management, CPU scheduling, and paging.
- Each group must submit the application, GitHub repository link, published website link, five-page PDF documentation, and presentation slides through the LMS.
- All four group members must contribute actively and participate in the live demonstration and presentation.
- All submitted work must comply with the University's Academic Integrity and Academic Misconduct Policy.

ASSESSMENT:

1. Case Study Scenario

A software development team has been asked to create an interactive application that helps users understand how Operating Systems manage computer resources.

Each group will act as a software development team and design an application that demonstrates important Operating Systems concepts through interactive simulations, calculations, and visualizations.

The application should allow users to enter data, run simulations, observe results, and compare different Operating Systems techniques.

2. Project Selection

Each group must select **ONE** of the following projects or propose another Operating Systems application with instructor approval.

No.	Project	Description
1	CPU Scheduling Visualizer	Simulate FCFS, SJF, Round Robin, and Priority Scheduling. Display Gantt charts, waiting times, and turnaround times.
2	Memory Management Simulator	Demonstrate FIFO and LRU page replacement. Visualize memory frames, page hits, and page faults.
3	Smart OS Resource Manager	Develop an interactive dashboard demonstrating process execution, CPU allocation, and memory management.
4	Deadlock Detection Simulator	Visualize processes, resource allocation, and deadlock detection or avoidance.
5	Process and Thread Visualizer	Demonstrate process states, transitions, and concurrent execution.
6	Disk Scheduling Simulator	Compare FCFS, SSTF, SCAN, and C-SCAN with disk-head movement visualizations.

Note: Every selected application must include functional components for resource management, CPU scheduling, and paging. The selected project determines the application's main focus.

3. Project Development

Step 1 – Plan the Application

Identify the problem, application objectives, main features, expected inputs, and outputs.

Step 2 – Develop Using AI-Assisted Coding

Use VS Code with an approved AI coding assistant to generate, modify, and improve the application through natural-language prompts.

Example Prompt:

Develop an interactive Operating Systems simulator using HTML, CSS, and JavaScript. Include process scheduling, memory management, user inputs, algorithm comparisons, and visual results. Create a responsive interface.

Step 3 – Implement the Application

Develop the application with working algorithms, interactive controls, calculations, and suitable visualizations.

Step 4 – Test and Improve

Verify the correctness of the application and improve any identified problems.

Step 5 – Publish the Application

Upload the completed project to GitHub and publish a working online version.

4. Assessment Questions

Part A – Application Development (10 Marks)

Question 1 – Resource Management (2 Marks)

Develop an application that uses the Operating System as a resource manager to support multiprogramming.

Question 2 – CPU Scheduling (3 Marks)

Implement suitable process scheduling algorithms for a given computing problem. Compare the results and select the suitable algorithm.

Question 3 – Memory Management and Paging (3 Marks)

Implement different paging techniques for a given computing system. Compare the results and select the optimal paging technique.

Question 4 – Application Integration and Testing (2 Marks)

Integrate the resource management, process scheduling, and paging components into one functional application. Test the application using different inputs and verify the correctness of the results.

Part B – Discussion (5 Marks)

Question 5 – Discussion (5 Marks)

During the live demonstration, each group member must participate in the discussion and answer the following questions: Each student will be assessed individually on their understanding of resource management, process scheduling, and paging.

No.	Discussion Questions	Marks
1	Explain how your application uses the Operating System as a resource manager to support multiprogramming.	1
2	Explain how you selected the suitable process scheduling algorithm for a given computing problem based on your application results.	2
3	Explain how you selected the optimal paging technique for a given computing system, working in teams and communicating the results.	2
	Total	**5**

ANSWER GUIDELINES:

Students are expected to:

- Select an Operating Systems project and identify its objectives and relevant Operating Systems concepts.
- Develop a functional application using VS Code and an approved AI coding assistant.
- Demonstrate resource management and multiprogramming concepts.

- Implement and evaluate suitable CPU scheduling algorithms.
- Implement and compare paging techniques.
- Test the application using at least three scenarios and verify results using manual calculations or trusted examples.
- Document at least five development prompts, their outputs, and the students' verification or modifications.
- Publish the source code to GitHub and deploy the application online.
- Submit exactly five pages of technical documentation, along with presentation slides.
- Demonstrate the application and answer technical questions individually.

Discussion Requirements (5 Marks)

- Each group member must participate in the discussion and answer questions individually.
- 1 mark: Explain how the application uses the Operating System as a resource manager to support multiprogramming.
- 2 marks: Explain how you selected the suitable process scheduling algorithm for a given computing problem based on your application results.
- 2 marks: Explain how you selected the optimal paging technique for a given computing system. Discuss how your team worked together and communicated the results.
- Use the developed application, simulation results, and comparisons to support discussion answers.
- Be prepared to demonstrate relevant application features during the discussion.
- All four group members must communicate their understanding during the discussion.

REFERENCING STYLE:

Students should use the American Psychological Association style (APA) as per the following guidelines: https://apastyle.apa.org/style-grammar-guidelines/references/examples

PLAGIARISM & GENERATIVE AI (FOR ASSIGNMENT BASED ASSESSMENTS):

Amity University Dubai is committed to academic honesty and integrity.
Generative AI is explicitly permitted for this assessment as a development assistant, including code generation, debugging, interface development, and testing support.

However, students must review, understand, test, and verify AI-generated code. Students remain responsible for the accuracy, originality, and academic integrity of their submissions.
Students must acknowledge AI assistance and external sources appropriately. Submitting unverified AI-generated work or claiming work they cannot explain as their own is not acceptable.

LATE SUBMISSIONS (FOR ASSIGNMENT BASED ASSESSMENTS):

You are advised to strictly adhere to the deadline. While Amity University Dubai might consider extenuating circumstances to support students who are experiencing unforeseen issues impacting their ability to complete assessments on time, any anticipated submission delay must be communicated in advance to the instructor who will consult with the Program Leader and the Dean before a permission is granted. All permissions must be based on valid reasons backed up by concrete evidence. For more details, you are advised to consult the latest version of AMUD's Academic Integrity and Academic Misconduct Policy.

FEEDBACK PLAN:

Formative feedback will be provided during laboratory sessions, project development activities, and office hours.
Students will receive feedback on application functionality, OS algorithm accuracy, prompt engineering, testing, documentation, and discussion preparation.
Summative written feedback will be provided on 22/12/2026, following the assessment evaluation.

ASSESSED COURSE LEARNING OUTCOMES:	RELATED QUESTIONS	MAX SCORE	MARK
CLO2: Use Operating System as a resource manager to support multiprogramming.	Part A – Resource Management (2 marks) + Integration and Testing (1 mark) + Part B – CLO2 Discussion (1 mark)	4	
CLO3: Select the suitable process scheduling algorithm for a given computing problem.	Part A – CPU Scheduling (3 marks) + Integration and Testing (1 mark) + Part B – CLO3 Discussion (2 marks)	6	
CLO4: Select the optimal paging technique for a given computing system, working in teams and communicating.	Part A – Paging (3 marks) + Part B – CLO4 Discussion (2 marks)	5	
	TOTAL	**15**	

EMBEDDED PLO DIRECT ASSESSMENT QUESTIONS? ☑ **YES** ☐ **NO**
[IF YES, COMPLETE THE TABLE BELOW]

PLO	EMBEDDED QUESTIONS
PLO1: Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	Application design, OS algorithms, resource management, scheduling, paging, and testing

PLO3: Communicate effectively in a variety of professional contexts.	Five-page documentation, individual discussion, and live demonstration
PLO5: Function effectively as a member or leader of a team, demonstrating leadership, collaboration and project management in achieving objectives related to the discipline.	Group development, individual contribution table, shared GitHub work, and discussion participation

SCORING RUBRIC			**LEVEL OF ACHIEVEMENT**			
	DESCRIPTOR	**WEIGHT**	**4 [87%-100%]**	**3 [73%-86%]**	**2 [60%-72%]**	**1 [0%-59%]**
1	Resource Management – CLO2	2	Accurate and functional resource-management features	Mostly correct features	Partial implementation	Incorrect or missing
2	CPU Scheduling – CLO3	3	Correct algorithms, results, and evaluation	Minor errors	Partial implementation or evaluation	Incorrect or missing
3	Paging – CLO4	3	Accurate implementation and comparison	Mostly correct	Partial implementation	Incorrect or missing
4	Integration and Testing – CLO2 (1 mark), CLO3 (1 mark)	2	Well-integrated application with verified tests	Minor issues	Limited testing	Nonfunctional or untested
5	Discussion – CLO2	1	Clear explanation of resource management and multiprogramming	Mostly correct explanation	Partial explanation	Incorrect or missing
6	Discussion – CLO3	2	Suitable scheduling selection supported by accurate results	Suitable selection with minor errors	Limited selection or justification	Incorrect or missing
7	Discussion – CLO4	2	Optimal paging selection, clear reasoning and team communication	Mostly correct selection and communication	Partial reasoning or communication	Incorrect or missing

Student Declaration

By signing this cover page, we acknowledge the assessment instructions and accept responsibility for complying with the University's Academic Integrity and Academic Misconduct Policy.
We declare that the submitted work represents our group's original work, that all external sources and AI assistance have been appropriately acknowledged, and that each group member has contributed to and understands the submitted application.

No.	**Student Name**	**Student ID**	**Signature**
1			

2			
3			
4			

Date: _____

To be filled by faculty

FEEDBACK FROM FACULTY:

Faculty Name and Signature **Date**